MagicBox Playbook Reference

Reference for every Ansible recipe in this repository: purpose, variables, commands, tags, prerequisites, verification and operational risks. It matches the code at the time of the build; `tools/check_docs.py` fails if a role, variable or tag is missing from this document.

Pinned toolchain: ansible-core 2.21.5 (Python 3.14 `.venv` managed by uv), community.general 13.5.0, community.docker 5.4.0, community.crypto 3.5.1.

1. Overview

Item	Value
Playbook	playbooks/site.yml
Inventory	inventory/hosts.yml (git-ignored; template inventory/hosts.example.yml)
Group	magicbox
Project policy	inventory/group_vars/magicbox/main.yml
Secrets	inventory/group_vars/magicbox/vault.yml (ansible-vault, git-ignored)
Collections	requirements.yml, installed to .ansible/collections
Result	Portainer CE at https://magicbox.local:9443/, LAN only

Play 1, *Configure MagicBox*, runs on group `magicbox` with `become: true` and applies the roles in this order: `preflight`, `mdns`, `firewall`, `docker`, `portainer`. Play 2, *Verify Portainer from the Ansible controller*, queries Portainer from the controller over the LAN.

2. Commands

Goal	Command
Create the environment	make setup
All local checks	make check
Container converge + idempotency test	make test-container
First deployment	make deploy EXTRA="-K -e magicbox_confirm_destructive=true"
Routine deployment	make deploy EXTRA="-K"
Only one area	uv run ansible-playbook playbooks/site.yml --ask-vault-pass -K --tags docker
Dry run with diff	uv run ansible-playbook playbooks/site.yml --ask-vault-pass -K --check --diff
Verification only	make verify EXTRA="-K"
Idempotency on MagicBox	make idempotency VAULT_PASSWORD_FILE=<file> EXTRA="--become-password-file=<file>"
List tasks / tags	uv run ansible-playbook playbooks/site.yml --list-tasks / --list-tags

`-K` asks for the sudo password; omit it if `<admin-user>` has passwordless sudo. A `--check` run cannot complete the `docker` and `portainer` roles on a fresh host because the packages they verify are not installed yet.

3. Tags

Tag	Selects
preflight	The preflight role
always	Also on preflight: it runs with any --tags selection, so checks and facts are always present
mdns	The mdns role
firewall	The firewall role
docker	The docker role
portainer	The portainer role, including TLS tasks

Tag	Selects
tls	Only TLS provisioning inside portainer (certificate renewal, CA fetch)
verify	Verification tasks in all roles plus the controller play
controller-verify	Only the controller play; skipped by the container test

`--tags verify` runs `preflight` (tagged `always`) and every `verify` task. It changes nothing on a converged host.

4. Shared variables

Defined in `inventory/hosts.yml` (environment facts) or `inventory/group_vars/magicbox/main.yml` (project policy).

Variable	Default	Description
magicbox_lan_interface	required	LAN interface on MagicBox (ip -br link). Avahi publishes only here.
magicbox_lan_cidr	required	Private IPv4 LAN subnet allowed to reach SSH, Portainer and mDNS.
magicbox_hostname	magicbox	Hostname and mDNS name (<name>.local); single DNS label.
magicbox_expected_architecture	x86_64	Asserted CPU architecture.
magicbox_supported_os	Ubuntu 24.04/26.04, Debian 12/13	Asserted OS/version map (ADR-0002).
magicbox_confirm_destructive	false	Allows destructive steps for one run (ADR-0008). Pass with -e.
portainer_admin_password	{{ vault_portainer_admin_password }}	Bootstrap admin password, from the vault.
vault_portainer_admin_password	required (vault)	At least 12 characters; in vault.yml only.

5. Role `preflight`

Purpose. Fail fast on unsupported or misconfigured hosts and prepare the baseline: OS/architecture/inventory/vault assertions, LAN CIDR validation, detection of where the SSH controller connects from, hostname, base packages.

Prerequisites. `magicbox.local` resolves from the controller (Avahi installed during the manual steps); SSH key login as a sudo-capable user; Python 3 on the host (present on Ubuntu/Debian server installs).

Variable	Default	Description
preflight_base_packages	ca-certificates, curl, gnupg, iproute2, python3-cryptography, python3-debian, python3-requests	Packages needed by later roles and modules.
preflight_manage_hostname	true	Set the static hostname and 127.0.1.1 entry. false only in the test container.

Facts set. `preflight_controller_location`: `inside`, `outside` or `unknown` (no `SSH_CLIENT`, e.g. the container test), consumed by `firewall`.

Verification. All assertions run on every play (tag `always`).

Risks. Changing `magicbox_hostname` renames the host and the mDNS name; certificates must match (`portainer_tls_dns_names` follows automatically, the existing CA's name constraints do not: rotate the CA, installation guide 8.5).

6. Role `mdns`

Purpose. Publish `magicbox.local` on the LAN with Avahi (ADR-0003). MagicBox has no fixed IP: Avahi re-announces the name whenever DHCP changes the address. Avahi is first installed by hand (installation guide 1.5) so Ansible can connect by name; this role then owns its configuration. Installs `mdns_packages`,

disables systemd-resolved's mDNS/LLMNR responders when systemd-resolved is installed, templates `/etc/avahi/avahi-daemon.conf` (host name, `allow-interfaces`, no HINFO/workstation publishing, no reflector, no wide-area), and enables `avahi-daemon`.

Variable	Default	Description
mdns_use_ipv6	false	Publish AAAA records. Off because the firewall only allows the IPv4 LAN.
mdns_packages	avahi-daemon, avahi-utils, libnss-mdns	Responder, CLI tools, NSS module for local resolution.

Handlers. *Restart systemd-resolved*, *Restart avahi-daemon*.

Verification. `avahi-resolve-host-name -4 magicbox.local` must return an address (retries 5 x 2 s).

Risks. Avahi renames the host to `magicbox-2.local` if another device already claims the name; TLS then fails the name check. Restricting to `magicbox_lan_interface` means the name is invisible on other interfaces.

7. Role `firewall`

Purpose. Enable ufw with default deny inbound and allow only SSH, `firewall_lan_tcp_ports` and `firewall_lan_udp_ports` from `magicbox_lan_cidr` (ADR-0004). Rules persist across reboots.

Variable	Default	Description
firewall_ssh_port	22	SSH port allowed from the LAN.
firewall_lan_tcp_ports	[9443]	TCP ports allowed from the LAN (Portainer).
firewall_lan_udp_ports	[5353]	UDP ports allowed from the LAN (mDNS).
firewall_logging	low	ufw logging level.

Safety rails. Refuses first activation unless `magicbox_confirm_destructive=true`; always refuses when `preflight_controller_location` is `outside`.

Verification. `sudo ufw status verbose` (manual); the controller play proves 9443 is reachable from the LAN.

Risks. A wrong `magicbox_lan_cidr` can lock out SSH from other subnets (console recovery: installation guide 8.2). Rules for ports no longer listed are not removed automatically; delete them with `sudo ufw delete <n>`. Ports published by containers on Docker bridge networks bypass ufw.

8. Role `docker`

Purpose. Install Docker Engine from Docker's apt repository with a verified signing key and a major-version pin, configure the daemon, and enable `containerd` and `docker` at boot (ADR-0002).

Variable	Default	Description
docker_major_version	"29"	apt preference pin 5:29.* for docker-ce, cli, rootless-extras.
docker_apt_key_url	https://download.docker.com/linux/<distro>/gpg	Signing key URL.
docker_apt_key_fingerprint	9DC858229FC7DD38854AE2D88D81803C0EBFCD88	Expected key fingerprint; mismatch aborts.
docker_apt_key_path	/etc/apt/keyrings/docker.asc	Keyring path referenced by the deb822 source.
docker_packages	docker-ce, docker-ce-cli, containerd.io, docker-buildx-plugin, docker-compose-plugin	Installed packages.
docker_conflicting_packages	docker.io, docker-doc, docker-compose, docker-compose-v2, podman-docker, containerd, runc	Removed only with confirmation.

Variable	Default	Description
docker_log_max_size	10m	json-file log rotation size.
docker_log_max_file	3	json-file rotated files kept.

Files. `/etc/apt/sources.list.d/docker.sources`, `/etc/apt/preferences.d/docker-ce.pref`, `/etc/docker/daemon.json` (validated with `dockerd --validate` before replacement; `live-restore` enabled).

Handlers. *Restart docker* (containers survive thanks to live-restore).

Verification. No `dockerd` TCP listener (`ss -Hltnp`).

Risks. Removing conflicting packages stops workloads that used them. A daemon restart briefly interrupts the Docker API. A major upgrade requires changing `docker_major_version` deliberately.

9. Role `portainer`

Purpose. Run Portainer CE with persistent data, HTTPS on 9443 using the MagicBox CA, plain HTTP disabled, and a pre-set admin password (ADR-0004, ADR-0005, ADR-0006). Container: `portainer`, image `portainer/portainer-ce:2.45.3`, restart policy `unless-stopped`, `network_mode: host`; mounts the Docker socket, the `portainer_data` volume, the certificate directory (read-only) and the password file (read-only).

Variable	Default	Description
portainer_image	portainer/portainer-ce	Image repository.
portainer_version	"2.45.3"	Image tag (LTS); also asserted against the API.
portainer_container_name	portainer	Container name.
portainer_https_port	9443	HTTPS port (keep in firewall_lan_tcp_ports).
portainer_data_volume	portainer_data	Persistent named volume mounted at /data.
portainer_base_dir	/opt/magicbox/portainer	Host base directory.
portainer_cert_dir	<base>/certs	TLS material (0700).
portainer_secret_dir	<base>/secrets	Admin password file (0700).
portainer_tls_mode	local_ca	local_ca or provided.
portainer_tls_dns_names	magicbox.local, magicbox	Certificate SANs and CA name constraints.
portainer_tls_ip_addresses	[]	Optional IP SANs/constraints. Keep empty: the IP is dynamic (DHCP).
portainer_tls_ca_common_name	MagicBox Local CA (magicbox)	CA subject CN.
portainer_tls_ca_valid_for	+3650d	CA lifetime.
portainer_tls_cert_valid_for	+397d	Server certificate lifetime.
portainer_tls_renew_before	+30d	Re-issue window before expiry.
portainer_tls_ca_fetch_dest	artifacts/magicbox-local-ca.crt	Controller copy of the CA for clients.
portainer_tls_provided_cert_src	empty	provided mode: server certificate path on the controller.
portainer_tls_provided_key_src	empty	provided mode: server key path (keep it vaulted).
portainer_tls_provided_ca_src	empty	provided mode: issuing CA path.

TLS tasks (`tls_local_ca.yml`). CA key (ECDSA P-384) and self-signed CA (`CA:TRUE, pathlen:0`, critical name constraints) created once; server key (ECDSA P-256), CSR and certificate (EKU `serverAuth`) signed by the CA and re-issued when within `portainer_tls_renew_before` of expiry or when the SANs change; CA fetched to the controller.

Handlers. *Restart portainer* when TLS material changes.

Verification. `GET https://magicbox.local:9443/api/system/status` from MagicBox with CA

validation (retries 20 x 3 s) and an assertion that the reported version equals `portainer_version`. The controller play repeats the request over the LAN.

Risks. Portainer controls the Docker socket and is root-equivalent: every Portainer admin is root on MagicBox. Changing the vault password after first start does not change the admin password. Deleting `portainer_cert_dir` creates a new CA that clients must trust again. Deleting the `portainer_data` volume erases all Portainer settings; the playbook never does this.

10. Testing the recipes

Check	Command	Proves
Syntax	make syntax	Playbook parses with the example inventory
Lint	make lint	yamllint + ansible-lint production profile
Secrets	make secrets-check	No plaintext vault values or private keys tracked
Docs	make docs-check	Links, ADR registry, this reference covers all roles/variables/tags
Container	make test-container	Full converge and changed=0 second run in an Ubuntu 24.04 systemd container (arm64 or amd64 per workstation)
Host	make deploy, make idempotency, make verify	Real behaviour on MagicBox

The container test overrides `magicbox_expected_architecture`, `magicbox_lan_interface`, `magicbox_lan_cidr` and sets `preflight_manage_hostname: false`; it cannot prove LAN mDNS or reachability from other machines.

11. Changing the recipes

1. Edit the role; keep variables prefixed and documented in `defaults/main.yml`.
2. Update this reference (roles, variables, tags, risks) and, for significant decisions, add an ADR and list it in the registry and `CLAUDE.md`.
3. `make manuals check test-container`.
4. Commit sources and rebuilt PDFs together.